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Mind Everywhere: A Framework for Conceptualizing Goal-Directedness in Biology and Other Domains—Part One

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Abstract
What makes a system—evolved, engineered, or hybrid—describable by teleological and mentalistic terms such as intelligent, goal-directed, cognitive, and intentional? In this two-part article, we review classical thought on teleology in the life sciences and defend a new approach to goal-directedness that stems from an emerging field—diverse intelligence. This field seeks to characterize what all active agents, regardless of their composition or provenance, have in common. Our approach emphasizes: (1) empirical testability (not philosophical commitments to linguistic categories), (2) fecundity in discovery of new capabilities (not just reductive mechanistic explanations of results after they are made, but worldviews that facilitate and enable novel research), (3) operationalization of terminology by reference to conceptual and empirical toolkits shown to be effective for a given system (cognitive and teleological claims are really hypotheses of optimal interaction protocols), and (4) continuity of human goal-directedness with our unicellular origins (which implies a need for models of scaling of cognition). In Part One, we review historical and contemporary debates about teleology in biology and describe some ground-breaking experimental results from morphogenesis, synthetic biology, and other areas of research involving the coordination of cellular behavior by means of bioelectric networks. In Part Two, we argue that current approaches, which tend to minimize the emphasis on goal-directedness in biology (and thus suppress the use of powerful tools from behavioral science), cannot account for these experimental results in a manner that helps us better understand how to control living systems toward new large-scale functions and fruitfully guide novel experimental work and practical applications in these areas of research. What is needed, therefore, is an approach to teleology in biology informed by mentalistic concepts, such as intelligence, cognition, and intentionality. We describe and explain our mentalistic approach in greater depth, show how it can be applied to specific cases, and address some important objections to our view. By abandoning the reductionistic aversion to mentalistic concepts in biology in favor of principled and testable frameworks for understanding diverse minds across a spectrum, embodied in the physical world, the plasticity and problem-solving competency of the agential material of life can provide fodder for advances in philosophical thought, as well as for biomedical and bioengineering applications.

Keywords Developmental biology · Teleology

Introduction: Teleology in Biology

Teleological concepts have played an essential role in biological explanation and discovery since the beginnings of scientific inquiry. Aristotle (384–322 BCE) held that to fully explain an event or process, one must understand not only its material, efficient, and formal causes but also its final cause or *telos*, that is, purpose (Aristotle [350 BCE] 2009b; Mayr 1982).[1] Purposes refer to the intentions, goals, or

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1 For an example of Aristotle's four causes consider the explanation of building a house. The wood, pipes, shingles, and other materials would be the material causes; the carpenters, plumbers, roofers, other

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plans of intelligent beings, and functions are what things do to serve their purpose. On this view, artifacts serve human purposes, whereas natural things serve divine ones. Moreover, the moral or aesthetic value of a thing derives from its purpose. For example, a good flute is good at performing its function (that is, to produce musical tones) and a good (or virtuous) person is good at performing the human function, which Aristotle understood to be rational activity of the soul (Aristotle [350 BCE]2009a).

Aristotelian ideas about scientific explanation and methodology held sway for nearly 2,000 years until philosophers and scientists, such as Galileo Galilei (1564–1642), Descartes (1596–1650), Robert Boyle (1627–1691), and Isaac Newton (1643–1727), developed and experimentally tested laws and theories that did not refer to goals or purposes. By the end of the 1700s, many scientists viewed the universe as a mechanical clock that operates according to mathematical principles of cause and effect (Jacob 2018). Though most scientists still believed that God created the universe, they saw no need to appeal to divine purposes to understand events and processes within the universe, such as planetary motion, the refraction and reflection of light, or chemical reactions. Interestingly, however, the least action principle of Pierre Maupertius (1698–1759) introduced a kind of primitive goal-directedness into the very center of physics (Annila 2011; Georgiev and Georgiev 2002; Ogborn et al. 2006)[2].

While teleological thinking soon faded from many aspects of astronomy, physics, and chemistry, it did not dissipate from biology, medicine, and natural history, because many scientists still accepted the theological viewpoint, which had been forcefully defended by William Paley (1743–1805), that the *only* reasonable explanation of the appearance of design in living beings is that it is due to the operation of a divine designer. However, in 1859 Charles Darwin (1809–1882) dealt what seemed to be a fatal blow to teleology in biology when he published the *Origin of Species*, a remarkable book that demonstrated, through rigorous arguments and copious evidence, that the appearance of design and order in the living world arises through random variation, adaptation, and differential reproduction and survival over geologic time (that is, natural selection) and not as the result of any divine plan or purpose (Darwin 1859; Dennett 1995; Mayr 1982). A key aspect of the conflict between these two worldviews is the assumption that there is a categorial difference (that is, a difference in natural kinds) between mechanistic and goal-directed systems. Before the establishment of cybernetics (Rosenblueth et al. 1943; Wiener 1948), it was simply not clear how any understandable system (that is, a machine) could have goals (Busseniers et al. 2021; Heylighen 2022; McShea 2013; Nechansky 2007).

Although biology has become increasingly mechanistic since the advent of Darwinism in the late 1800s and the rise of genetics, biochemistry, biophysics, and molecular biology in mid-1900s, teleological thinking persists in biology, because describing biological phenomena in terms of goals or functions is a useful—and some would say inescapable—way of thinking about the structure and behavior[3] of cells, tissues, organisms, and other biological systems (Allen and Neal 2020; Monod 1972; Rosenberg 1985).[4] To paraphrase a famous quote from evolutionary biologist Theodosius Dobzhansky (1900–1975), nothing in biology makes sense without teleology (Dobzhansky 1973; Levin 2023b; Toepfer 2012).

Despite its methodological value in biology, teleological thinking has faced numerous attacks over the years (Allen and Neal 2020; Mayr 1992). Francis Bacon (1561–1626) argued that explanations of scientific phenomena that refer to goals are useless expressions that do not advance our understanding of nature or make successful predictions (Bacon 1605). However, it is often the case that knowing that a biological system has a particular goal can provide us with useful information. For example, if we have seen little black ants (*Lasius niger*) in our kitchen, we can predict that if we leave an uncovered piece of cake on our kitchen countertop overnight, it will likely be covered by black ants by the morning. We can also predict they will take larger food objects back to the nest by rotating them through obstacles in very clever ways (Dreyer et al. 2025), and that their collective behavior can be fooled by geometric illusions that also fool vertebrate nervous systems (Sakiyama and Gunji 2013, 2016).

A more fundamental objection to using teleological language in biology is that it violates the metaphysical axiom that causes must precede their effects (Nagel 1979; Pearl et al. 2016; Ruse 1973; Wright 1976). Saying that "an acorn germinates in order to become an oak tree" seems to imply

builders would be the efficient causes; the blueprints would be the formal cause; and the end goal (the house itself) would be the final cause.

[2] Thus, these least action principles answer the question of "how far down does agency go?"—it goes all the way to the bottom and reveals the minimal lower bound on the spectrum of agential goal-directedness.

[3] We will use the term "behavior" in a very broad sense in this article to include any form of activity associated with life, including but not limited to movement, energy utilization, metabolism, waste production, development, growth, reproduction, and evolution (Gómez-Márquez 2021).

[4] For a clear example of the role teleological thinking plays in biological discovery, consider inquiry into the chemical structure of DNA (Resnik 1995). In constructing their model of DNA, James Watson and Francis Crick (1916–2004) were guided by assumptions concerning its biological function and purpose, for example, "what must the chemical structure of DNA be like to enable it to serve the goals of information storage and be self-replication?"

that an effect (that is, the goal of becoming an oak tree) exists prior to a cause that brings it about (that is, germination). Although interactions at the quantum level of reality may sometimes violate the temporal order of causes and effects, interactions above this level do not (Wolchover 2021). This is a problem for simple systems, where the most explanatory story that can be told refers only to immediate causes; Still, for more complex ones, especially biological systems, it is often the case that the most effective models for prediction and control emphasize the anticipatory or even counterfactual information that guides behavior[5] (Friston 2013; Friston and Ao 2012; Rosen 1979, 1985).

Quantum physics aside, the most straightforward way of circumventing the backwards causation problem is to interpret teleological language mentalistically; that is, to understand goals as mental representations (such as desires, wants, intentions, or preferences) that guide behavior (Woodfield 1976).[6] For example, if we observe a person folding a piece of paper into an origami crane, we might explain their behavior by saying that they have a goal of making the crane, which guides their behavior. In this case, there is no problem with backwards causation because the mental representation of the crane exists prior to the actions that create the crane. Woodfield (1976) is one of the few contemporary philosophers to defend a mentalistic approach to teleology in science.[7] Woodfield argues that notions of purpose and function apply primarily to purposive behavior in human beings and intelligent animals and that applications beyond this realm are metaphorical or derivative.[8]

A key problem with the mentalistic approach, according to many scientists and philosophers, is that it involves ascribing mental properties to things that clearly *do not* have minds (Bedau 1992; Wright 1976). While most scientists will agree that it makes sense to say that human beings and other intelligent animals have minds that allow them to represent goals and think about how to seek them, they will not say that it makes sense to say the same things about cells, plants, insects, swarms of bees, or developing embryos.[9] A paramecium that swims toward a food source is assumed not to *think about* where it is going; it simply acts on the basis of stimuli. To think otherwise is often seen as succumbing to the delusion of anthropomorphism.[10]

Another objection to the use of teleological language in biology is that it is thought to imply a commitment to mysterious "vital forces" linking biological goals to their effects (Rosenberg 1985). Although modern biologists resoundingly reject vitalism because it is incompatible with materialist metaphysics, past prominent biologists and philosophers have defended this doctrine (Mayr 1982). For example, the German embryologist Hans Driesch (1867–1941) argued that one cannot adequately explain morphogenesis without assuming that it is guided by vital forces that control cell differentiation, growth, and movement (Driesch 1914). French philosopher Henri Bergson (1859–1941) argued that living things have a vital force that can produce creativity in evolution (Bergson 1911). A key question of this field now is to show how the apparent entelechy can be compatible with the facts of chemistry and physics, much as a non-physical "algorithm" can be causal in making the electrons dance inside a computer circuit despite their obeyance of Maxwell's equations (Auletta et al. 2008; Ellis and Drossel 2019; Ellis 2008, 2009, 2012; Ellis and Bloch 2011; Ellis et al. 2012; Walker et al. 2017).

Although these objections have generated considerable philosophical and scientific discussion, they have not convinced most biologists to abandon teleology. Additionally, since the 1950s, prominent biologists and philosophers, including Ernest Nagel (1901–1985), Ernst Mayr (1904–2005), Jacques Monod (1910–1976), Colin Pittendrigh (1918–1976), Francesco Ayala (1934–2023), and many others, have helped to reduce the cognitive dissonance that can arise from using teleological language in biological inquiry by showing how to interpret teleology in a way that does not imply backwards causation, vitalism, or anything else that conflicts with scientific materialism. An essential part of this rapprochement is the idea that mentalistic approaches to teleology apply *only* to the behavior of human beings

5 Indeed, it is a hallmark of agential systems that an effective explanation of what they are doing, and an effective strategy to manipulate their behavior, must involve events in a larger "cognitive light cone" around them than the local here and now. Local stimuli are sufficient to deal with a billiard ball, but events at a spatial distance, in the past or in the future, become crucial components of effective models of agents with memory, predictive capacity, and a radius of concern that is bigger than their immediate surface (which is already true for cells, and certainly for multicellular creatures).

6 We will say more about mental representation later in this article. For now, we will say that we are justified in treating a system as if it has mental representations insofar as this assumption helps us to fruitfully describe, interpret, explain, predict, or control the behavior of the system (Dennett 1987).

7 See also Ducasse (1925).

8 Of course, there is nothing inherently suspect about metaphors in science, since they are used all the time in the physical sciences for explaining and describing phenomena. For example, electrons are described as "jumping" between orbitals or forming a "cloud" around the nucleus. In immunology, antibodies attach to antigens like a "lock and key."

9 We note that radical behaviorists, such as Skinner (1971) do not accept the use of teleological language even to describe human behavior.

10 Of course, the charge of anthropomorphism is a compact way of smuggling in the prescientific notion that humans have a kind of magic inaccessible to other systems; this neglects our continuity with single cells (developmentally and evolutionarily). The project of modern teleology is not to confer human-style magic upon others but instead to naturalize human capacities and explain how the capacities attributed to humans have scaled up from minimal origins.

and some intelligent animals (Allen and Neal 2020; Bedau 1992; Mayr 1992; Woodfield 1976).

Here, we argue that confining mentalism in this way is a mistake: it is an unnecessarily limiting view that creates pseudo-problems by positing sharply delimited natural kinds that are not supported by state-of-the-art biology and bioengineering (Clawson and Levin 2022). It contributes to a kind of teleophobia that inhibits powerful research programs in regenerative medicine and synthetic morphoengineering.[11] Recent advances in the fields of basal cognition and diverse Intelligence (Baluška and Levin 2016; Lyon 2006, 2015, 2020) show that it is useful to view many different biological systems, including molecular networks, cells, tissues, and organs—as well as organisms—as acting intelligently. By *intelligence (*or *goal-directedness)* we mean *the ability to achieve goals by different means* (focusing on problem-solving in action spaces, including but not limited to the familiar 3-dimensional space of conventional "behavior") (Levin 2022).

The degree of intelligence of a system falls on a spectrum (or relative scale), depending on the extent to which a system satisfies criteria associated with functional goal-directedness (Fig. 1). Specifically, these criteria can be objectively demonstrated by the empirically tested degree of efficacy of different sets of tools applied to the system, ranging across physical rewiring, cybernetics/control theory, behavioral science, linguistic commands, reasoning, and psychoanalysis. At one end of the spectrum are highly intelligent agents, such as human beings, which demonstrate many different capacities related to the pursuit of goals, such as memory, learning, problem-solving, planning, self-awareness, metacognition, and language. Specifically, they have metacognitive loops that allow them to reason about their own goals and reset some of them. At the other end of the spectrum are simpler agents, such as genetic regulatory networks, which demonstrate only a few cognitive capacities, such as exploratory plasticity and learning (Levin 2022, 2023a, 2023b 2023c; Levin and Dennett 2020), and below that the least action behavior of passive matter[12]. Somewhere between these two ends of the spectrum one may find agents with various goal-directedness traits, such as nonhuman mammals, developing embryos, ant colonies, and artificially intelligent machines and hybrids of biological and technological components (Levin and Dennett 2020; Rouleau and Levin 2023). A key task of biology, then, is to determine the position of various systems on this spectrum, since the types of interventions that may be effective at controlling these systems will vary, depending on where they are on the spectrum (Fields and Levin 2022). This determination must be arrived at empirically, by making hypotheses about action space, goals, and competencies to reach those goals, and then doing experiments to verify and improve our theory of mind about these systems—not by holding on to ancient philosophical commitments about which systems can (or cannot) be described in cognitive terms.

Strong recent support for our position comes from experimental work in morphogenesis, synthetic biology, and other fields that study bioelectric networks outside the brain, some of which we describe in the section on "New Experimental Results in Morphogenesis, Synthetic Biology, and Related Fields Involving Coordination of Cellular Behavior via Bioelectric Networks" (Lagasse and Levin 2023; Levin 2019, 2023b). Some of the new discoveries about the plasticity of cell and tissue systems, and new ways to prompt them toward complex anatomical outcomes, were made possible by a framework that implies that groups of cells act as agents in pursuit of collective goals and that the tools of computational neuroscience and behavioral science can be applied outside of their normal domain of brainy organisms navigating 3D space (Levin 2023a). Although we maintain that our view is philosophically defensible, our main concern in this essay is to develop a position that has scientific and practical value; that is, an approach that yields testable, predictive, and explanatory hypotheses that lead to new ways of thinking in biology and have practical implications in medicine, bioengineering, robotics, and other fields (Levin 2012, 2019, 2021a, c, 2022; Levin and Dennett 2020). Finally, we also would like to stress that nothing in our view should be taken to imply vitalism, divine design, or other doctrines that involve a commitment to mysterious forces or phenomena that are incompatible with empirical science or limit rational inquiry and progress (Fields and Levin 2022; Levin and Dennett 2020).

In Part One of this article, we review historical and contemporary debates about teleology in biology and describe some groundbreaking experimental results from morphogenesis, synthetic biology, and other areas of research involving the coordination of cellular behavior by means of bioelectric networks.

[11] Dresow and Love (Dresow and Love 2023) refer to this philosophical aversion to talk of minds, purposes, intentions, and goals in nature as "teleophobia."

[12] Even there, an engineer finds a degree of autonomy that can be exploited. For example, when building a roller coaster, one provides a mechanism to get it up the hill, but there is no mechanism needed to make sure it gets back down—it's an autonomous competency of the system to do that. Minimal, true, but not zero, and we hold a component of the capabilities (free energy minimization (Allen and Friston 2018; Kirchhoff et al. 2018; Kuchling et al. 2020; Ramstead et al. 2018, 2019) that when scaled up results in full-blown human metacognitive capacities.

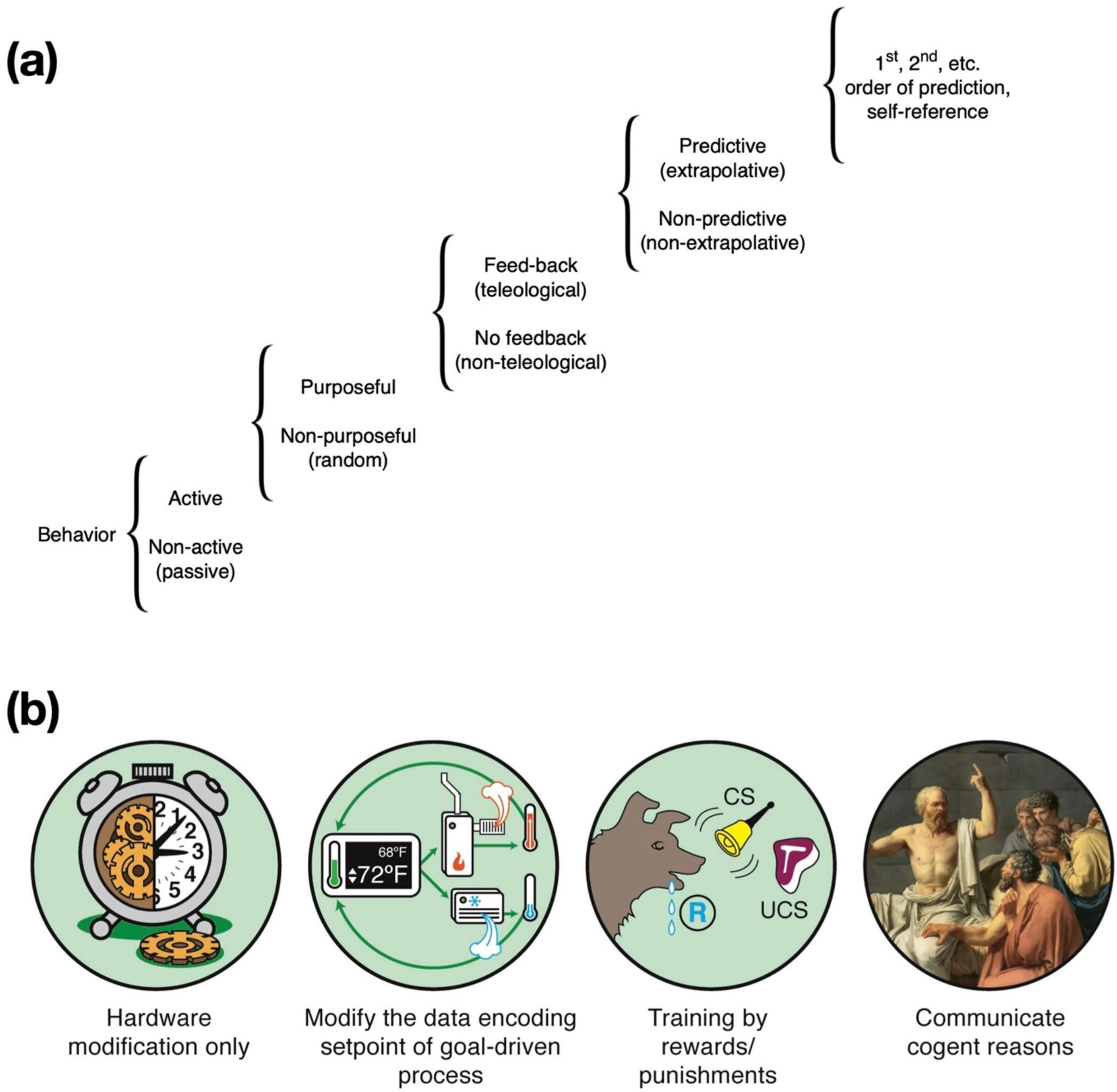

Fig. 1 The continuum of cognitive properties. **a** A set of behavioral competencies, leading from passive matter to high-grade metacognitive intelligences (such as humans), illustrating great transitions in information processing from the cybernetic perspective. Taken with permission from (Rosenblueth et al.1943). **b** A spectrum of persuadability (Levin 2022), which illustrates the concept that cognitive terms are interaction protocol claims: the cognitive status of any system, from simple mechanical devices that only follow least action laws to ones that can be predicted and controlled by the tools of cybernetics, behavior science, or psychoanalysis, can only be ascertained by function experiments in which one uses the tools appropriate to a given type of diverse intelligence and gauges the empirical efficacy of the interaction. Image in (b) by Jeremy Guay of Peregrine Creative Reprinted with permission.

Brief Review of the Contemporary Debate About Biological Teleology

Teleology Versus Teleonomy

Although biologists and philosophers, including Immanuel Kant (1724–1804), Jean-Baptiste Lamarck (1744–1829), and Darwin have debated about the meaning of function and design in nature since Aristotle's time, contemporary inquiry into to the problem of biological teleology began in the 1950s, when Pittendrigh (1958) and Mayr (1961, 1974, 1992) introduced the distinction between *teleology* and *teleonomy* as a way of talking about goals and functions in biology without presupposing mentalism, vitalism, or divine design (Dresow and Love 2023). While both terms refer to goal-directedness, one way of distinguishing between teleology and teleonomy has to do with whether a system itself has metacognition about its own goals, or whether the goals are only apparent to an outside observer. A strongly teleological system has components that can process information about (and perhaps reset) its own goals—in a sense, it knows it has goals. In a more teleonomic system, an outside observer can determine that it has goals (by perturbational experiments that attempt to deviate the system from its goals), but no evidence suggests that the system is in any way representing or regulating its own goals. For example, a person who folds a sheet of paper to produce an origami crane is a teleological system because they have mental access to the represented goal (that is, the goal of making the crane) that guides their behavior and does it with second-order intent to pursue a goal under different conditions. For example, the person could continue making the crane even if they cut their finger and need to bandage it. In contrast, a 1st-generation thermostat relentlessly making sure the room is at the correct temperature might be merely teleonomic because, while it effectively pursues a goal, it lacks the apparatus to represent facts about goal-having or to implement the stress mechanisms that motivate effort and plasticity when goals aren't being achieved, to change the setpoint according to larger-scale meta-goals, and so on. In any case, determining the degree to which a system has true teleology, especially for unconventional and nonlinguistic systems (or even linguistic ones, in the cases of robotically embodied large language models), is not trivial and requires empirical experimentation.

Mayr (1961, 1992) argued that a teleonomic system must have an internal program (such as the genome) that produces goal-directed behavior. However, as we shall see below, there are many different approaches to teleonomy (Dresow and Love 2023). These approaches can be divided into two groups: ahistorical accounts, which hold that purposes, goals[13], and functions can be defined without reference to their history; and historical (or etiological) accounts, which hold that purposes and functions can be defined only by referring to their history (Garson 2016). While the distinction between teleology and teleonomy has historical importance, it is not widely used in biology, despite some recent efforts to revive it (Dresow and Love 2023)[14]. Therefore, in the remainder of this essay, we will use the term "goal-directedness" to refer to the phenomena that have been described as teleological or teleonomic.

Ahistorical Theories of Goal-Directedness

Building on the work of Sommerhoff (1950) and Braithwaite (1953), Nagel (1961, 1979) defended an ahistorical account of goal-directedness based on concepts from cybernetics, the study of communication and control systems in animals and machines (Wiener 1948). Nagel argued that two cybernetic notions are crucial for understanding goal-directedness in any system: persistence and plasticity. Persistence is the tendency for a system to produce or maintain a pattern of behavior despite external perturbations that could make it deviate from this pattern. Plasticity is the tendency of the system to produce a pattern of behavior from different starting points (McShea 2012). For example, a thermostat in a house exhibits persistence because it maintains a selected temperature in the house despite fluctuations in heat affecting the house. A thermostat exhibits plasticity because it can achieve the temperature setting from different starting points, that is, initial temperature readings in the house. Nagel argued that one cannot infer goal-directedness only from observing the *behavior* of the system; one must also have evidence that the system has homeostatic *mechanisms* that produce goal-directed behavior (Garson 2016). For example, a thermostat exhibits persistence and plasticity because it has homeostatic mechanisms, including heat sensors, heating system control switches, and feedback loops, that enable it to control the temperature of the house.

A key assumption made by Nagel is that statements about functions and goals should explain *why something exists or persists* in a system. Hempel (1965) and Cummins (1975) criticized this view on the grounds that inquiring about the function of something does not necessarily tell us why it exists or persists. The statement "the function of the mammalian heart is to pump blood" does not tell us precisely why mammals have hearts, because there are many different types of blood-pumping mechanisms that could perform this function, such as three-chambered pumps, multiple pumps, and so on. The term "heart" stands for a set of functional

[13] We can define goals as concrete states within a problem space, which a system expends effort to achieve (with some observable degree of competency despite barriers, interventions, and other novel scenarios). We can define purpose as a goal that a given system can reason about (metacognition)—the system represents itself as a goal-driven system and often can work to change those goals in accordance with a more general ensemble of meta-goals.

[14] It should be noted, however, that there is another use of the word teleonomy (Levin 2023b), which is not meant to diminish its reality or importance, but rather to emphasize (via the word "apparent") the nature of an observer from whose vantage point the teleological behavior can be detected and exploited. This is the perspective of polycomputing (Bongard and Levin 2023) and more generally of observer-focused frameworks (Fields et al. 2022; Fields and Levin 2022) (and of course, complex agents are also observers of themselves, grounding their own teleonomy).

properties that can be realized in many different physical systems (Rosenberg 1985).

Based on this, as well as other criticisms of Nagel's view, Cummins (1975) developed a *causal role theory of goal-directedness*. The basic insight behind this theory is that functional statements explain *what things do*, not why they exist (Amundson and Lauder 1994; Couch 2023). Complex biological systems have many different, interacting parts at various levels of organization. The parts of the system have capacities to produce or maintain various effects (or goals) within the system (Eronen and Brooks 2024; Wimsatt 1972a). For example, the human body has eleven different organ systems, composed of 78 organs, 206 bones, 600 muscles, 200 cell types, and 30 trillion cells (Marieb and Hoehn 2018; Sender et al. 2016).[15] To understand the function of any part in this system, one must determine what it does in the system and how it interacts with other parts. For example, macrophages are immune system cells that carry out several *functions* that help to promote the *goal of protecting the body against disease*. One of these functions is to phagocytize and degrade dead cells and foreign materials. Macrophages also interact with other immune system cells, including T-cells and B-cells, to protect the body against disease. Macrophages can activate T-cell responses by presenting them with antigens from pathogens or cancer cells and by secreting cytokines (Guerriero 2019). Macrophages are equipped with proteins that help them perform their functions, including receptor proteins that recognize molecular patterns of dead cells and foreign materials, enzymes that facilitate phagocytosis, and cytokines. These proteins are synthesized by ribosomes from genetic information encoded in the DNA (Lendeckel et al. 2022).

Like Cummins, McShea (McShea 2012) also regards information about biological complexity and hierarchical organization as essential to the understanding of goal-directedness, but he criticizes the causal role theory because it still involves statements about function and goals within systems and therefore risks sliding back into mentalism. McShea (2012, 2023), Lee and McShea (2020), and Babcock and McShea (2021) have developed an ahistorical view that explains persistence and plasticity in biological systems in terms of hierarchically organized structures, or fields, rather than goals and functions. A field can provide direction and control for entities affected by it. For example, a magnetic field can align iron filings in a particular position or deflect the solar wind. To illustrate the idea of field in biology, McShea (2012) describes bacteria swimming toward a food source dissolved in water. The concentration gradient of the food in the water guides the bacteria to the source, not the goal of reaching the food. Moreover, the bacteria exhibit persistence and plasticity with respect to moving toward the food because they will move in the direction, irrespective of perturbations in the system or different starting points. According to McShea, one can (in theory) understand almost all biological systems, including ant colonies, human societies, and ecosystems, in terms of fields and interactions between fields (Babcock and McShea 2021).

Finally, it is important to mention Newman's (2023, 2024) notion of biological functions as properties inherent in biological systems, especially multicellular organisms, specifically eschewing adaptationist focus on selection for reproductive fitness. According to Newman and his collaborators (2025), certain types of biological functions arise because of morphological and physiological enablements for the organism-as-agent, which actively invents new ways of living and harnessing environmental affordances. Examples of such include the ability of collections of cells to form and repair basic structures, such as spheroids and layers; to move via cytoskeletal structures, such as cilia; to divide; to form chemical gradients; and, depending on the type of species, to form more complex structures, such as cell walls and appendages (Newman 2023, 2024; Newman et al. 2025). These properties arise naturally in certain types of living matter and extend to chemical systems such as phospholipids that have the ability to form barriers in water because the hydrophobic ends of these molecules turn away from the water, and the hydrophilic ends turn toward the water. This ability plays a key role in cell structures because it enables cells to form membranes that are used to sequester the internal contents of the cell from the external environment. While Newman et al. (2025) do not focus on mentalistic terminology, we think their approach to agency in evolution of form and function is completely compatible with our efforts to make use of specific concepts from cognitive and behavioral science to understand the details of how the organism-as-agent (at all scales, from molecules to whole body and perhaps beyond (Tung et al. 2023) finds solutions and projects its agency into new spaces.

Historical Theories of Goal-Directedness

A key problem with the ahistorical approach to goal-directedness in biology, according to its critics, is that it does not provide sufficient context to distinguish between functions, accidents, and dysfunctions (Garson 2016, 2017; Millikan 1984, 1989; Neander 1991; Ruse 1973). For example, the human heart produces sounds, which a doctor can listen to for diagnostic purposes. A heart murmur can indicate a serious problem, such as aortic stenosis (Mayo Clinic Health

[15] Even something as "simple" as a yeast cell is composed of about 42 million protein molecules, consisting of 5,858 different types of protein (Ho et al. 2018); 6,000 genes; and 200,000 ribosomes (Woolford and Baserga 2013). Additionally, cells can learn (Gershman et al. 2021; Kukushkin et al. 2024) and anticipate future events (Pershin et al. 2009; Saigusa et al. 2008).

Letter 2000). Even though producing sound is a useful effect of the heart, physicians would not say that this is its function. It is, at best, a fortuitous accident that heart sounds can be used in medical diagnosis, since the heart's function is clearly to pump blood. Ahistorical approaches cannot support the obvious conclusion that the heart's function is not to make sound, since the heart is structured and organized so that it exhibits persistence and plasticity in producing this effect. With respect to the function/dysfunction distinction, a normal, adult human heart beats at 55–85 beats per minute at rest (Lewine 2023). If a person has a resting heart rate of 130 beats per minute, this would be regarded as unhealthy or dysfunctional. However, it is difficult to substantiate this claim without referring to contextual information, as their heart may be organized and structured to exhibit persistence and plasticity in producing this effect.

Proponents of historical theories of goal-directedness argue that history provides the context that is needed to distinguish between functions and accidents and functions and dysfunctions. For example, a historical theorist could claim that the function of the heart is to pump blood, as opposed to making sound, because pumping blood contributed to the adaptive fitness of human beings with hearts, whereas making sound did not. A historical theorist could also say that a human heart that beats 130 times per minute at rest is not functioning properly because this effect would have had a negative fitness.

Historical theories of goal-directedness in biology have been defended by Ayala (1970); Wimsatt 1972b; Ruse (1973); Wright (1976); Boorse (1977); Brandon (1981); Millikan 1984, 1989; Neander (1991); Garson (2017, 2019). Most proponents of the historical approach claim that biological functions are phenotypes or genotypes that persist in populations because they contributed to the fitness of the organisms that possessed them. That is, they are adaptations (Brandon 1981). Thus, these historical accounts explain *why* something exists by referring to *what* it did in the past, that is, how it affected fitness. However, some theorists (for example, Millikan 1989; Garson 2017, 2019) generalize this point and argue that any historical process that selects properties based on their effects can produce functions. For instance, if a horse learns to avoid wire fences after receiving several shocks from a wire fence, one could say that the function of its wire fence-avoidance behavior is to protect the horse from electric shocks.

Critics of historical theories of goal-directedness argue that selection history provides neither necessary nor sufficient conditions for ascribing functions or goals to biological phenomena. It does not provide sufficient conditions because genotypes or phenotypes that are adaptations to past environments may no longer serve a function in their current environment (Garson 2016, 2019). For example, the hemoglobin beta (HB) mutation is common in populations that live in regions where malaria is endemic, such as parts of sub-Saharan Africa and the Middle East. The mutation changes the shape of the hemoglobin protein and causes sickle cell disease in homozygotes but not heterozygotes. Since heterozygotes have a copy of the normal allele, they can produce normal red blood cells and do not develop sickle cell disease, but they still have resistance to malaria. So, according to theories that define functions in terms of natural selection, the function of the mutation is to provide malaria resistance (Sabeti 2008). However, this mutation no longer serves that purpose in populations that now live in regions where malaria is not endemic. Historical theories do not provide the necessary conditions for ascribing functions because some traits may contribute to reproduction and survival even though they have not yet been favored by natural selection. For example, suppose a mutation emerges in a population that confers resistance to an infectious disease, such as COVID-19, but has not been in the population long enough to be favored by natural selection. It seems plausible to claim that the function of the mutation is to provide resistance to COVID-19, but the historical view cannot countenance this hypothesis until the mutation persists and is selected for (Bigelow and Pargetter 1987).

Pluralistic Theories of Goal-Directedness

Some authors have responded to the debate between ahistorical and historical theorists by arguing or suggesting that neither view entirely captures the meaning of goal-directedness in biology and that the most reasonable way of thinking about this concept is to adopt a pluralistic approach (Sober 1993; Griffiths 1993; Godfrey-Smith 1993; Brandon 2013; Garson 2016, 2018). That is, "purpose," "function," and related terms can mean different things in biological research, depending on one's epistemic aims and the types of questions being posed (Salmon 1984). When the primary objective it to understand *how* something works (for example, when doing research in comparative anatomy or physiology), the ahistorical view may be most apt; however, when the primary objective is to understand *why* something exists or persists (for example, when doing research in evolutionary biology or ecology), the historical view would seem to be most useful. In many types of biological inquiry, both types of questions will arise, and answers to one type of question may suggest answers to the other (Garson 2016, 2019). For example, information about brain evolution can be useful in understanding the brain's neural architecture and vice versa (Verendeev and Sherwood 2017).

Table 1 Theories of goal-directedness in biology. Theories of goal-directedness in biology can be classified as mentalistic theories, which explain purposes and functions in terms of mental categories, such as mind, cognition, and problem-solving; and non-mentalistic theories, which do not explain teleology in terms of mental categories. Non-mentalistic theories can be subdivided into ahistorical theories, which explain the purpose or function of a thing in terms of its current causal role or place within a physical-mechanical system, and historical theories, which explain the purpose or function of the thing in terms of its history. Pluralistic theories combine historical and ahistorical accounts. Although we defend a mentalistic approach in this article, our view could be characterized as pluralistic and pragmatic because we are not rejecting these other approaches to goal-directedness and believe that the appropriateness of any approach depends on its ability to fruitfully guide experimentation and inquiry.

Mentalistic theories	Non-mentalistic theories
Aristotle ([350 BCE] 2009a, 2009b)	*Ahistorical Theories*
Woodfield (1976)	Homeostatic mechanisms (Nagel 1961)
Levin (2022)	Genetic program (Mayr 1992)
Dennett (1987)	Causal roles (Cummins 1975), (Amundson and Lauder 1994)
Deacon (2012)	Physical or chemical fields (McShea 2012)
Juarrero (2015, 2023)	Inherent properties of biological systems discovered by embryo-as-agent; (Newman 2023; Newman et al. 2025)
Dresow and Love (2023)	*Historical Theories*
	Natural selection; Ayala (1970), Ruse (1973), Wright (1976), Brandon (1981)
	Generalized selected effects; Millikan (1989), Neander (1991), Garson (2017,2019)
	Pluralistic Theories
	Sober (1993), Griffiths (1993), Godfrey-Smith (1993), Brandon (2013), Garson (2016, 2018)

Brief Interlude

As one can see from the synopsis in previous section, the leading theories of goal-directedness in biology fall roughly into three camps: the ahistorical approach, which interprets statements about goal-directedness as offering explanations of *what* things do or *how* they work; the historical approach, which interprets these statements as offering explanations of *why* things exist or persist; and pluralist approaches which regard both types of interpretations as acceptable, depending on scientific usage. These theories are non-mentalistic because they attempt to show how teleological terms can be understood without referring to minds or mental states. Although these different theories yield some useful insights into teleological thinking, it is not clear to us that any of these theories has gained widespread acceptance among biologists.[16]

While we think it is important to review these leading approaches to biological goal-directedness to help the reader better understand our view, it is not our aim in this article to adjudicate this debate. This is an *important project*, but it is *not our* project (see, for example, Garson 2016; Wouters 2005). Rather, the main objective of this article is to advance a framework for thinking about goal-directedness in biology that is in some ways a revival of an older, mentalistic approach. The rationale for this new approach comes not from philosophical analysis but from the empirical success of an alternative framework that has facilitated new experimental work in morphogenesis, synthetic biology, and bioelectric networks, which has wide-ranging implications for thinking about goal-directedness in biology and other domains. A new approach is required because non-mentalistic theories of goal-directedness do not provide an adequate explanation of these findings, nor do they offer fruitful guidance for future work in these and other areas involving the organization of complex behaviors.[17] Before developing our view further, we will describe examples of experimental work in morphogenesis and synthetic biology that support our view and explain why mainstream approaches to biological goal-directedness fail to provide optimal guidance for further progress in these fields.

New Experimental Results in Morphogenesis, Synthetic Biology, and Related Fields Involving Coordination of Cellular Behavior via Bioelectric Networks

Every result in experimental biology can, after it is discovered, be said to be implemented by mechanical molecular processes underneath. The same is, of course, true of the most advanced, uncontroversial examples of human goal-directedness (Sapolsky 2023). And this is not surprising: the mechanism is always going to be chemistry (or quantum foam, if one is truly committed to the reductionist project), never fairy-dust. However, we propose that a hallmark of a good philosophical framework is that it facilitates the discovery of new capabilities and engenders new research programs, not just that it explains events after they've been produced. Here, we briefly describe some specific discoveries that were made precisely because of the use of a teleological framework that allowed the empirical testing of agential toolkits (primarily, from the cognitive and behavioral sciences) on novel substrates from which they were previously isolated due to the gatekeeping of allegiance to prescientific categories. Many more can be found in recent work (Baluška et al. 2010, 2016; Baluška and Mancuso 2009; Bentley and Chakravartula 2017; Bentley et al. 2014; Brigandt and Love 2012; Bugaj et al. 2017; DiFrisco et al. 2023; Dresow and Love 2023;

[16] During the 1970s, the term "teleonomy" started to gain favor, but teleology is back in vogue (Dresow and Love 2023).

[17] This situation in biology is similar, we believe, to what happened in physics during the early 20th century, when the double-split experiment required physicists to rethink their theories of light and matter (Allori 2015).

Gremiaux et al. 2014; Kelz and Mashour 2019; Koseska and Bastiaens 2017; Koshland 1980, 1983; Love 2010; Martinez-Corral et al. 2019; McFadden and Koshland 1990; Moczek et al. 2011; Noble 2010, 2011, 2012, 2021, 2022a, 2022b; Prindle et al. 2015; Sultan et al. 2022; Yang et al. 2020; Zakirov et al., 2021).

One example of the practical implications of specific hypotheses about the intelligence of a given system is revealed by work on frog craniofacial morphogenesis. Every tadpole needs to rearrange its face to become a frog (Kanamori and Brown 1996; Shi et al. 1996; Tata 1996). It was assumed that this is a hardwired process—every tadpole looks the same and every frog looks the same. Hence, all evolution had to do was provide for mechanisms that reliably moved every organ in the right direction by the right amount. We (that is, Dr. Levin's lab) decided to empirically test this assumption (Vandenberg et al. 2012), suspecting that, in fact, the system may have more intelligence than that. Context-sensitive problem-solving competencies need to be revealed by perturbational experiments that probe what kind of goal-directedness a system might have, in addition to open-loop (feed-forward) emergence of complex outcomes from the rote execution of low-level mechanics. Thus, we created what we called "Picasso tadpoles," where the craniofacial structures were scrambled—the eye might be on top of the head, the mouth might be on the side, and so on. It was found, using quantitative morphometrics (Pinet et al. 2019; Vandenberg et al. 2012), that in fact these scrambled animals still made quite normal frogs, because the organs would go through novel paths (indeed, sometimes going too far and needing to double back) to end up in the correct frog face target morphology. This example shows the ability of morphogenesis to reach a specific goal despite unexpected starting conditions (a more sophisticated way of navigating the anatomical morphospace), and more generally, the benefits of not assuming a system is at the very low end of problem-solving capabilities but of testing to see what behaviors it can muster to meet its normal end-state when circumstances change.

An even more drastic example can be seen in regenerating planaria. Planarian flatworms are amazing creatures—they regenerate every part of their body and can be cut into many pieces, each of which will form a perfect little worm (Saló et al. 2009). Of course, this process is studied today using conventional molecular biology and models of pathways (Owlarn and Bartscherer 2016). However, there is an interesting aspect that is not predicted or implied by the standard paradigm. Planaria are not only highly regenerative, but also cancer-resistant and ageless (asexual planaria do not age and are basically immortal). Because they reproduce by fission and regeneration, they accumulate somatic mutations. For 400+ million years, every mutation that does not kill a stem cell is amplified into the next generation as the cells rebuild new bodies after fission. They are mixoploid, with different numbers of chromosomes in their bodies (discussed in Levin et al. 2018). Of course, this can be studied using molecular methods, but note that the modern paradigm of the genome as a mechanical orchestrator of biological events does not predict that the animal with the messiest genome should be the one with the most robust regeneration abilities and resistance to cancer and aging. It's the exact opposite prediction one would make from a genome-as-driver view. An explanation for this bizarre mismatch between expectation has finally been offered and simulated in computational experiments (Shreesha and Levin 2023), and it relies on the *morphogenetic competency* of the cellular material: by conceptualizing evolution as producing problem-solving agents with the ability to achieve specific goals in anatomical morphospace (Levin 2023c), the situation becomes clear. An unreliable material leads to selection for computational capability to achieve goals despite the presence of noise. Planarian tissue, due to the somatic inheritance of mutations, is ultimately unreliable and exerts significant pressure to evolve highly competent decision-making circuits to reach the normal target morphology despite novel circumstances. This gives rise to a feedback loop, because such regulative processes hide information from selection (a good embryo might be good because the structural genome is great, or because it wasn't great but the embryo fixed the problem on the fly, as often occurs (Levin 2023c). This, in turn, means that evolution spends more time tuning the competency capability and not the baseline, default hardwired morphogenetic mechanisms.

One other interesting aspect that this line of thinking explains is the lack of transgenics in planaria. In other model systems, one can obtain mutant lines—fruit flies with curly wings, mice with unusual coat colors or skeletal malformations, and so on. In planaria, not only is there no transgenesis available (despite 30+ years of attempts), but there are no mutant lines! Why? On the mechanistic view, this is a major puzzle. However, it makes perfect sense if one uses the concept of cell groups as a collective intelligence that solve anatomical problems (reviewed in Fields and Levin 2022; McMillen and Levin 2024): planaria have strongly de-emphasized their genetics, and instead leverage enormous autonomy and morphological intelligence (this is also seen in their ability to solve novel physiological problems (Emmons-Bell et al. 2019). Interestingly, there are two permanent lines of planaria that are not normal—but they are made by perturbing not genetic mechanisms, but the same kind of bioelectric signaling that underlies advanced animal cognition (Durant et al. 2017, 2019; Oviedo et al. 2010; Pezzulo et al. 2021).

Planarian fragments can be induced to form two heads instead of one; the first two-headed forms were made at the turn of the last century (Child 1911; Morgan 1898). However, to our knowledge, no one had *re-cut* the two-headed forms to see what would happen until our laboratory did it in 2009. Likely, this was because it was perfectly obvious what would happen: the genetics are normal, and therefore fragments would build what they always build, going back to one head. Our framework instead led to a different prediction (Fig. 2). First, we reasoned that the head–tail decision could not be local or determined by morphogen gradients (see Levin et al. (2018) for explanation) but must be a collective distributed decision. Electrical synapses are a powerful mechanism for ensuring collective decision-making in

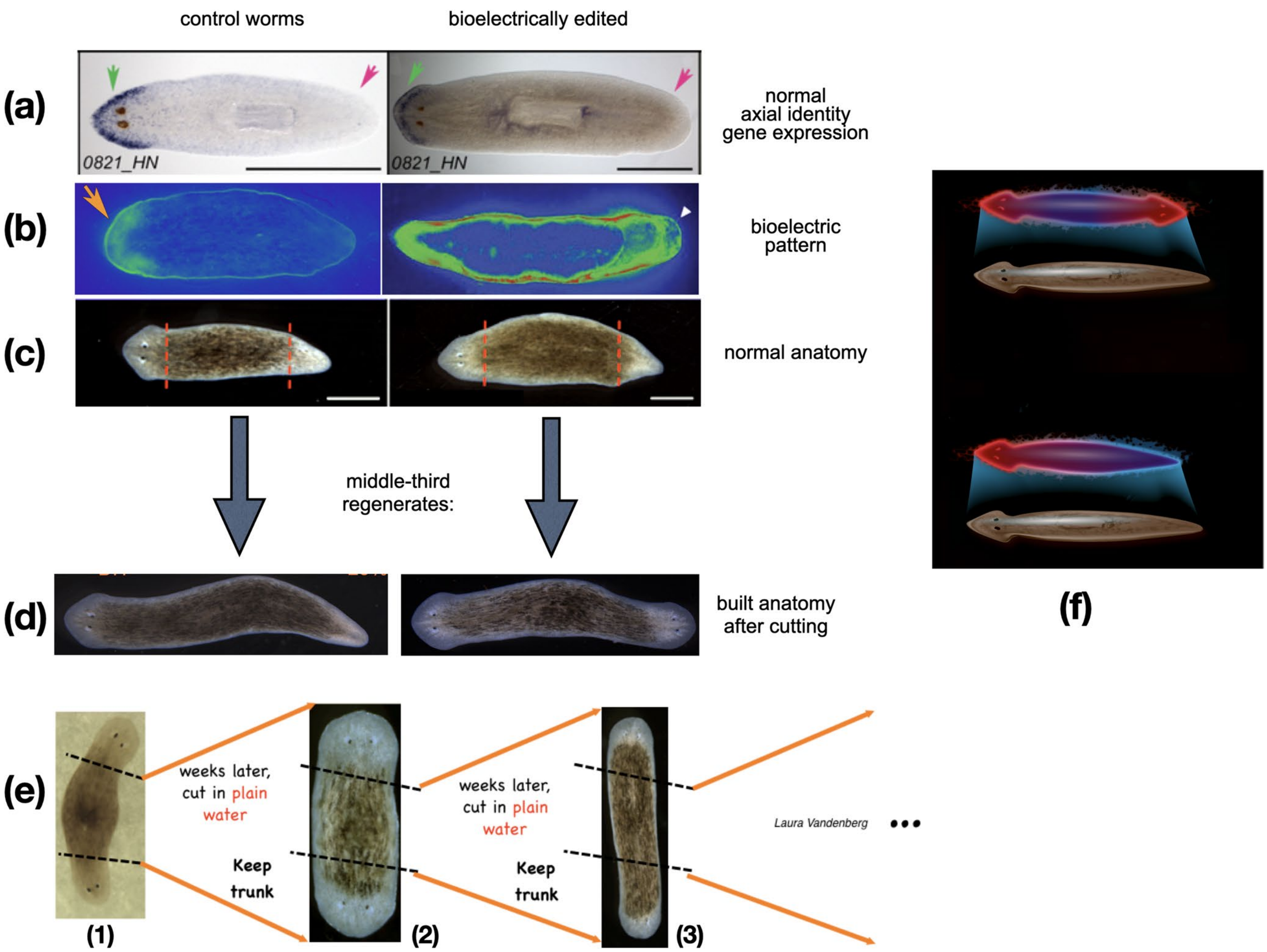

Fig. 2 Rewriting morphogenetic goals in regenerating planaria. Flatworms (in this case, *Dugesia japonica*) regenerate after amputation. The left columns in panels **a–d** show control worms, while the right panels show worms in which the endogenous bioelectric prepattern has been altered by exposure to a drug that alters the resting potential of cells (Durant et al. 2019; Pezzulo et al. 2021). **a** Planaria have head-identity genes (blue stain, in situ hybridization with an anterior marker) expressing in the correct side (anterior, not in the posterior tail region). **b** Voltage imaging with a fluorescent reporter dye (Oviedo et al. 2008) reveals the bioelectric pattern in which depolarized regions (green, left panel) specify that there should only be one head in normal worms. In a drug-manipulated worm (right panel), the pattern has been altered to encode the "two heads" outcome. Both animals have normal anatomy—one head (oriented to the left), and one tail (on the opposite side). Red dashed lines indicate the plane of amputation. **d** When those animals are cut, the middle fragments regenerate the number of heads consistent with the bioelectric pattern they had. **e** When two-headed animals are amputated further, with no more exposure to bioelectric reagents, pieces continue to regenerate as two-headed across generations (1, 2, 3) without limit. These data show that a bioelectric pattern stored in the tissue serves as the target to which the cells will build when needing to restore a whole body and that the pattern which determines this is not stored in the genetics—the same hardware can implement multiple outcomes (and stop when the outcome is achieved), as schematized in **f** where the same physical planarian body is shown storing multiple different representations of what the correct state is, which will guide repair and remodeling. The anatomical setpoint can be reset without changing the hardware of the system (as in any homeostatic system with a memory of goal state) and is persistent, though re-writable (a key property of memory). Panels a-e taken with permission from Durant et al. (2019); Durant et al. (2017); Oviedo et al. (2010); Pezzulo et al. (2021). Panel f by Jeremy Guay of Peregrine Creative.
Reprinted with permission (Full-color figure available in electronic version)

the brain (Palacios-Prado and Bukauskas 2009). Thus, we tested their roles in planaria, discovering that two-headed worms could be made by perturbing the electrical synapses known as gap junctions (Nogi and Levin 2005). Then, reasoning that this collective dynamic is likely to have the kind of memory seen in neural collectives (albeit in morphological space, not behavioral 3D space), we re-cut the two-headed worms, discovering that the two-headed state is *permanent*—in the absence of any further manipulations, two-headed worms' fragments continue to regenerate as two-headed (Oviedo et al. 2010). This is a remarkable example of nongenetic inheritance of a different large-scale anatomy in a multicellular organism with a wild-type genome.

We were also able to target the bioelectric circuits storing the number of heads to be made (Durant et al. 2019) (a literal encoding of a goal for the collective intelligence of cells—a biophysical mechanism of representation of the state toward which all the cells will work upon injury). Not only could we reset it back to one-headed (Beane et al. 2011), but we could also produce what in neuroscience is called perceptual bistability—an indecision and stochastic oscillation between two different interpretations of the same input (Pezzulo et al. 2021): we created worms whose fragments stochastically produce one or two heads (Durant et al. 2017). A key aspect of this work is that what we rewrite is a stable bioelectric state within tissue. When we modify this pattern memory, it stays latent: a one-headed form, with normal anatomy and molecular marker expression, can be given a bioelectric pattern memory of "two heads." The memory and anatomy stay discordant until and unless it is injured, in which case the cells use this memory to know what to build (constructing two-headed worms). In this sense, it is a truly counterfactual memory—it does not represent what is true about the animal's body now, but what *will* be true if it gets injured, and in an important functional sense, the activity of the cells is controlled by an internal representation of a future goal-state. *That is, the teleology of the system is implemented by having an encoded state of the future (not backward causation at the level of physics, but at the level of information) that guides the morphogenesis toward a specific, encoded end.*

These experiments demonstrate how an unconventional system (working in anatomical morphospace) can have a representation (or encoding) of a future state via the cellular collective of the body. It may represent an evolutionary precursor to the amazing "time travel" capacities of brains, which can think about situations that do not currently exist. This kind of representation of future or past goal states is an important component of concluding that a system has a degree of mind (without necessarily assuming meta-cognition or subjective experience, that is, qualia). Indeed, if we were to try to visualize the basal forms from which complex mammalian counterfactual representation might have evolved, this simple scheme would be the kind of thing we would imagine as the basic mechanisms that nervous systems later elaborated, amplified, and projected into the world of 3D behavior. The example also shows a specific, experimental approach to directly visualizing the encoded goal representations of unconventional systems, because the bioelectric voltage dye imaging reveals the (re-writable) goal state to an external observer, much as neural decoding is supposed to give 3rd-person access to human mental goals and memories. Here, we know the goal exists because we can *see it* and *rewrite* it, thus moving the question from a purely philosophical discussion of goals to a practical, empirical scenario.

Note what this example demonstrates: by developing tools to observe (bioelectric dye imaging) and rewrite (targeting the ion channels in specific ways) the bioelectric pattern memories that serve as the future goals of the morphogenetic process, we reify the existence of goals, memories, and counterfactual states in an unconventional substrate. The best way to identify and prove the nature of a teleological system is by uncovering the ways in which goals are encoded (that is, represented) and rewriting them, enabling the system to achieve novel behaviors not because its genetically-specified hardware was micromanaged to force it (like a mechanical clock) but because it was given a novel goal on which to exercise its autonomy (like a sophisticated thermostat circuit with multiple ways of maintaining a set temperature). We know these things exist, not metaphorically but in the most real sense of all, because using the tools and concepts of neuroscience, we facilitate the ability to enable discovery and reach new capabilities by techniques used at the bench, which were not facilitated by conventional approaches hyper-focused on bottom-up mechanistic causes. This research is just one example of the broader project of exploring bioelectric networks (and the capacities implemented by them) as a kind of cognitive glue that binds together active subunits, from neurons to bacteria (Martinez-Corral et al. 2019; Prindle et al. 2015), toward the creation of emergent large-scale agents that have goals, competencies, memories, preferences, and top-down causal control than none of their parts have (Pezzulo and Levin 2015, 2016).

Another example of the benefits of an engineering approach to teleology (goals as aspects of the materials' autonomy that can be targeted in effective interventions) concerns our attempts to use the bioelectric interface to communicate novel goals in vertebrate development (Fig. 3). The face in frog embryos is defined by an instructive bioelectric prepattern that determines the location and number of eyes, mouth, and other organs (Vandenberg et

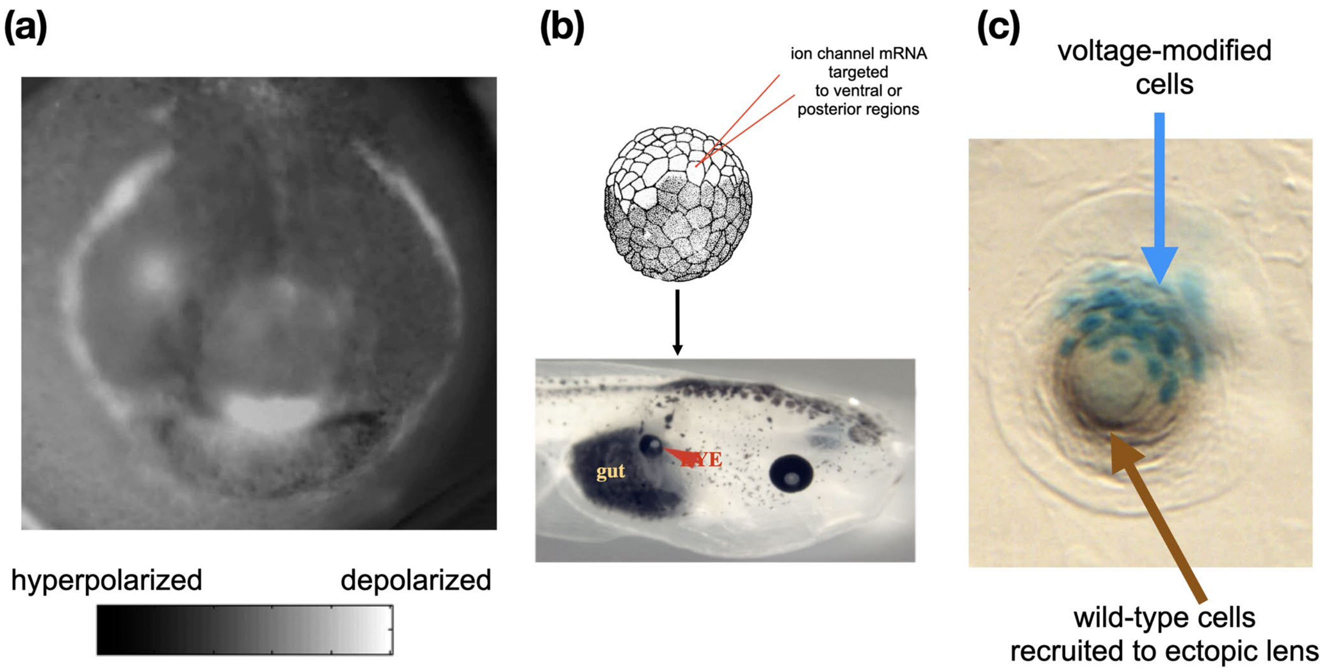

Fig. 3 Resetting organ-level goals for cell groups in vivo. **a** Normal craniofacial development in frog embryos is preceded by a bioelectric prepattern in which the borders between depolarized (light) and hyperpolarized (dark) regions set the locations of the various organs of the face and head. We asked whether these bioelectric prepatterns were instructive for the behavior of cell collectives. **b** In order to reproduce the eye spot elsewhere in the body, we injected an early embryo with mRNA encoding a potassium ion channel, which would establish a specific bioelectric pattern in the posterior of the embryo. The result was eye formation (red arrowhead) in regions such as the gut. Those eyes contained lens, retina, optic nerve, and other components, even though all we provided was a very simple "make an eye here" prompt (a voltage state), not detailed instructions on how to micromanage the gene expression and cell differentiation needed to actually make an eye. **c** When only a few cells were targeted (shown by the blue stain), they spontaneously recruited their unmodified neighbors to help in the task of making an ectopic lens in the animal's flank. This illustrates the many competencies of the material: modularity, interpretation of simple stimuli via complex behavioral programs that are self-organized, scaling of resources appropriate to the task, re-specification of collective goals on-the-fly, and so on. Panel a is reprinted with permission from Vandenberg et al. (2011). Other panels reprinted with permission from Pai et al. (2012) (Full-color figure available in electronic version)

al. 2011). When a similar pattern indicating the eye spot is introduced into other areas of the body, such as the gut, the cells build an ectopic eye (Pai et al. 2012). This can even happen in posterior regions, which, according to standard developmental biology, are incompetent to form an eye (this conclusion was reached because prior studies attempted to force eye development with the "master eye gene" *pax6*, which indeed doesn't work posterior to the neurectoderm at the front of the head; but the more convincing bioelectric prompt works everywhere). Note several key aspects which are facilitated by the hypothesis that cells are an agential material amenable to the tools of behavioral science—communication, goal-resetting, and collaboration—not just molecular micromanagement. First, it shows the bioelectric pattern memories are instructive, not epiphenomenal—they are causal for changes in complex morphology not easily reachable by conventional means. Second, we did not force the construction of an eye by managing gene expression and painstakingly orchestrating the position of all the cell types. We didn't need to do this any more than the humans of 5,000 years ago needed to know molecular mechanisms of neuroscience to communicate with dogs and horses. We provided a high-level prompt—a simple trigger stimulus which led to downstream complex behavior and the system managing its own molecular events to reach a goal in anatomical space that we specified (a dynamic well familiar to students of animal behavior. Third, targeting only a few cells is enough—they will autonomously recruit others, not initially targeted by us, to help complete the task. This, of course, also occurs in other collective intelligences like ants and termites, which recruit conspecifics to help with a heavy carrying task. Taken together, these aspects illustrate how taking seriously the autonomy and competency of the material confers empirically verifiable, practical advantages. Work is currently ongoing to explore other aspects of neuroscience in these morphogenetic contexts, including the use of anxiolytics (Shreesha and Levin 2024), hallucinogens (Sullivan and Levin 2016), serotonergic modulators (Levin et al. 2006), memory blockers, and many other ways of modulating decision-making, perception, and memory in full-blown teleological agents in novel substrates to show the utility of this approach for novel discoveries.

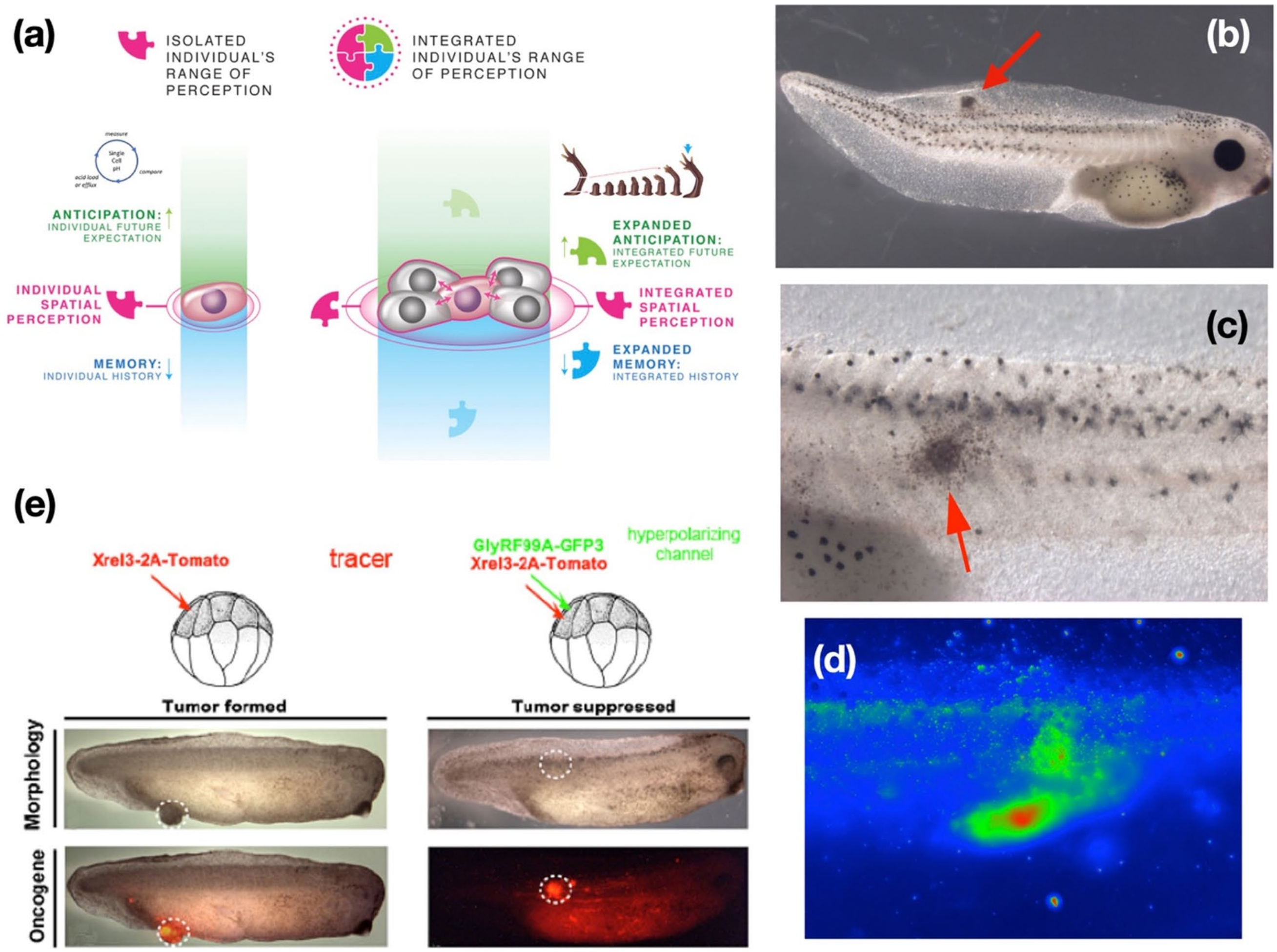

Fig. 4 Cancer as a dissociative identity disorder of the collective morphogenetic intelligence. **a** Multicellular morphogenesis can be described as the enlargement of the goal states, from individual cells' modest homeostatic goals about pH and other values within the current cell boundary, to cell groups pursuing grandiose goals (such as making sure a salamander limb has the right shape and size). These are goal states in the sense that they guide the activity of cells and determine when they stop (when the error has been reduced sufficiently). This process of connecting cells into higher-order computational networks that can store and pursue larger goals in new spaces (anatomical space instead of the landscape of metabolic, physiological, and transcriptional goals) has a failure mode: cancer. **b** When cells are disconnected from their collective by stress or oncogenes (such as the human oncogene injected into the frog embryo, which makes a tumor, red arrow), they revert back to their unicellular goals—a shrinkage of the boundary between self and world in which the cells no longer perceive their membership in a larger collective—a problem analogous to dissociative identity disorder in cognitive systems (Levin 2021b). **c** A close-up of the metastasizing tumor. **d** This bioelectric disconnection is apparent using voltage reporter dyes, which indicate the abnormal bioelectric state of the nascent cancer cells. **e** The use of this unconventional perspective on the cancer problem led to a specific treatment strategy: co-inject ion channels into the site of oncogene function to force the correct bioelectric state: whereas the oncogene alone causes a tumor (morphology in top left panels, fluorescent tracer on the oncoprotein in bottom panels), it is prevented from doing so when co-injected with an ion channel mRNA—there may be no tumor (top right panel) even though the oncoprotein is strongly expressed (red tracer, bottom right panel). This shows that cancer is not simply a hardware disorder due to mutation but is fundamentally tractable using tools that target the size of cellular goal states (Full-color figure available in electronic version)

Another example comes from cancer research (Fig. 4): whereas the planarian and eye induction examples showed rewriting the content of the represented morphogenetic goal, the following shows how the *scale* of cellular goals can be usefully manipulated. The bottom-up mechanical approach holds cancer to be an irrevocable result of genetic mutations spread as clonal populations from a single mutated cell (hardware defects). This has a practical implication (toxic chemotherapy to kill the cells), and a prediction: animals with ready access to proliferative, undifferentiated cells should be most prone to cancer. However, this prediction is wrong—the animals with the highest plasticity (planaria, salamanders) have a very low incidence of cancer (reviewed in Chernet and Levin 2013a; Moore et al. 2017). An alternative model has been proposed (Levin 2019, 2021c): cancer as literally a dissociative identity disorder of the somatic collective intelligence. Cells which electrically disconnect from the tissue whose network stores very large morphogenetic goals (build a limb, or a liver, and so on) in effect roll back to the tiny cognitive light cones[18] of single-cell amoebas, whose goals are metabolic, proliferative and migratory (that is, metastasis). This view suggests an alternative to chemotherapy and a possibility that is not reachable from the conventional assumptions: that cancer can be reversed not by fixing the molecular hardware, but by rescaling the *goals* of the system, enabling the cells to work towards larger goal states (organ maintenance). We tried this, and showed that: (1) carcinogenic defection can be seen early by tracking the bioelectrical states of cells (Chernet and Levin 2013b), (2) metastatic melanoma can be induced in the absence of carcinogens, oncogenic mutations, or DNA damage—simply by disrupting the cellular bioelectric communication (Blackiston et al. 2011; Lobikin et al. 2015a, b), and (3) cancer induced by potent oncogenes can be normalized into healthy tissue by forcibly re-establishing the correct bioelectric state of the collective, including at long range within tissue (Chernet et al. 2015, 2016; Chernet and Levin 2014). In parallel with resetting the goals discussed in the section above, this research program reifies goal-directedness by showing that the *scale* of goals is a useful, practical target of interventions that enables biomedically relevant progress.

The final example concerns the ability of unconventional biological agents to find *new goals*. The examples above showcased examples of exploiting the goal-directedness of the agential material of life (Davies and Levin 2023). But where do these goals come from? Some are shaped by evolution, the way that inborn instinctual behaviors are provided for some animals. But others are creatively developed in real-time, based on circumstances, as seen in many other examples of animal behavior. The conventional view sees morphology and behavior as driven by genetics and shaped by a long history of selection. We pursued the idea that cell collectives are not only goal-seeking agents but can acquire novel behaviors that do not require eons of shaping to a specific environment (that is, that evolution produces problem-solving agents, not fixed solutions). First, by liberating epithelial cells from the other cells of a frog embryo (which normally induce them to have a boring life as the two-dimensional outer surface of an embryo), we found that they spontaneously assemble into self-motile little constructs we dubbed *Xenobots*, which have novel capacities such as kinematic self-replication—the ability to make copies of themselves by building with cells found in the environment (Blackiston et al. 2021, 2023; Kriegman et al. 2020, 2021). To our knowledge, there is no other creature that reproduces by kinematic self-replication, and there have never been any Xenobots or selection to be a good Xenobot. They possess a highly altered transcriptome (Pai et al. 2024), despite the lack of any novel genes, synthetic biology circuits, scaffolds, or anything else—this is revealing the competencies and capabilities of the cell collective, and supports the view of the genome as a resource into which living beings dip as needed for their lifestyle, versus a prescription of what and how to be. Next, while the conventional paradigm held that this was an unusual property of amphibian embryonic cells, we explored the implications of the framework that sees plasticity as a fundamental aspect of minimal agents and investigated the capabilities of *adult human* cells. This led to the discovery of *Anthrobots*—self-motile proto-organisms made of adult human tracheal epithelial cells, which, among other things, have the ability to heal wounds in surrounding neural tissue (Gumuskaya et al. 2023). Likely, this is just the beginning, and we have argued that complementing the field of synthetic biology (attempting to build functionality bottom-up), bioengineering, and regenerative medicine can both take advantage of the competency and autonomy of the material (Lagasse and Levin 2023; Levin 2024; Mathews et al. 2023; McMillen and Levin 2024).

[18] The cognitive light cone (defined in (Levin 2019) demarcates the scale of the largest goal, in both space and time, which a system is capable of working towards.

Preliminary Conclusion

This concludes the first part of our new approach to goal-directedness in biology and other domains. In the second part, we will argue that current approaches, which tend to minimize the emphasis on goal-directedness in biology (and thus suppress the use of powerful tools from behavioral science) cannot account for these experimental results described in previous section in a manner that helps us better understand how to control these systems toward new large-scale functions and fruitfully guide novel experimental work and practical applications in these areas of research. What is needed, therefore, is an approach to teleology in biology informed by well-defined, rigorous, theory-backed mentalistic concepts, such as intelligence, cognition, and intentionality (and their attendant conceptual tools from neuroscience). In the second part, we will also describe and explain our mentalistic approach in greater depth, show how it can be applied to specific cases, and address some critical objections to our view.

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Declarations

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